

platforms like Heroku to create geo-distributed networks, and stream data into machine analytics platforms like Splunk.

OpenIoT: This is application-centric middleware for pulling information from sensors. It acts as Sensing-as-a-Service for deploying and providing services in a cloud environment.

Other open source middleware platforms are ThingsBoard, NATS, DSA, MainFlux, OpenIoT, Sitewhere, OpenRemote, EclipseIoT, etc.

Open source databases for analytics

The entire IoT implementation revolves around the idea of data collection/insertion through sensors and sending control back to devices. Selecting the right database in the various stages of IoT implementation for a given use case depends on multiple factors like enterprise adoption, business goals, as well as the development and operational perspectives.

The following are some of the top open source databases available for IoT based applications.

MongoDB: This is a powerful, flexible, document-oriented and scalable NoSQL database. It supports features like range queries, sorting, aggregations and geospatial indexes. It supports JSON to store and transmit information. Being a standard protocol, JSON is a great advantage for both the Web and the database. MongoDB supports a rich query language for read and write operations (CRUD) as well as data aggregation, text search and geospatial queries.

MySQL: This is an open source relational database management system that brings data consistency, scalability, high performance, availability and flexibility to IoT solutions by the efficient collection of data from IoT devices. It helps in data transformation through annotation and aggregation,

helping users understand the data in a better way.

Redis: This in-memory open source database is a popular choice for IoT solutions. It is widely used in IoT solutions for data ingest real-time analytics, messaging, caching and many other use cases. Today, it helps in edge computing involving deep learning, image recognition and other innovative computing requirements.

Cassandra: This is a highly scalable and distributed open source database for managing voluminous amounts of structured data across many commodity servers. It provides availability, linear scale performance, simplicity and easy distribution of data across multiple database servers. It supports strong data consistency across a distributed architecture.

Hadoop: This is an open source software platform for distributed storage and the distributed processing of very large data sets on computer clusters built from commodity hardware. It helps in driving analytics from all IoT data. It easily ingests data from multiple data sources, and supports both batch as well as real-time data ingest from sensors using tools such as Apache Kafka and Apache Flume. It handles multiple IoT data types, structures and schemas, and supports real-time processing and applications on streaming data. Hadoop is a flexible, scalable and secure database.

Other open source IoT databases are CrateDB, GridDB, InfluxDB, BigChainDB, etc.

Open source IoT communication protocols

IoT protocols are a crucial part of the IoT technology stack. Without them, hardware would be rendered useless as the IoT protocols enable exchange of data in a structured and meaningful way. IoT primarily exploits standard protocols and networking technologies.

The top five open source IoT communication protocols in use are described briefly below.

CoAP (Constrained Application Protocol): CoAP is an Internet application protocol for constrained devices. It has been designed for IoT systems based on HTTP protocols. CoAP makes use of the UDP protocol RESTful architecture. It is used within mobiles and social network based applications.

MQTT: This messaging protocol is used for remote monitoring in IoT. Its primary task is to acquire data from many devices, and transport it to the IT infrastructure. MQTT connects devices and networks with applications and middleware using hub-and-spoke architecture. The MQTT protocol provides efficient information routing functions to small, cheap, low-memory and power consuming devices in vulnerable and low bandwidth based networks.

XMPP: This is a communications IoT protocol for message-oriented middleware based on the XML language. It enables real-time exchange of structured yet extensible data between any two or more network entities. XMPP enables messaging applications to attain authentication, access control, hop-by-hop and end-to-end encryption.

AMQP: This is an application layer protocol that supports reliable communication via message delivery assurance primitives like 'at-most once', 'at least once' and 'exactly once'. The AMQP protocol consists of a set of components that route and store messages within a broker service. This protocol enables client applications to talk to the broker and interact with the AMQP model.

Data Distribution Service (DDS): This is used for real-time machine-to-machine communication. It enables scalable, real-time, dependable, high-performance and interoperable data exchange via the